

FOSS4GIS 2026

Spatial Risk Assessment of Heavy Metal Pollution in Serbian Surface Waters

GIS · Multi-Criteria Decision Analysis · Spatial Statistics · QGIS

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Why This Matters



More data, not more answers

Serbia collects water quality data from dozens of stations every year.

But having numbers and understanding what they mean spatially are two very different things.



Where exactly is the problem?

Standard monitoring tells us if a station crossed a limit. It does not tell us where risk is concentrated, how it spreads, or whether it is getting worse.



Pass/fail is not enough

A simple yes/no answer misses gradients between stations, changes over time, and how pollution moves from one part of a river system to another.

LEGEND

- Monitoring locations
- Hydrographic network
- Administrative districts of the Republic of Serbia

0 25 50 75 100 km

Monitoring Stations

Heavy Metals

Years (2019–2023)

Cadmium

Highly toxic, accumulates in kidneys over time. One of the most tightly regulated metals under the EU Water Framework Directive.

Lead

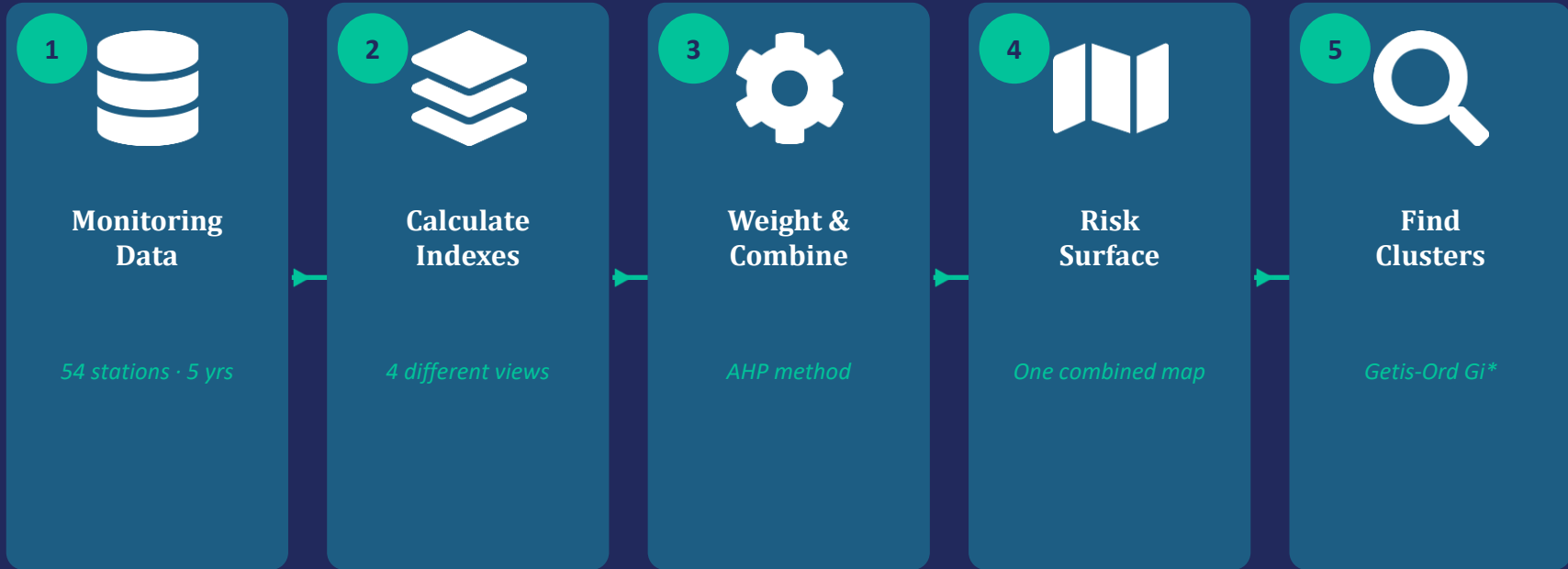
A potent neurotoxin that binds to river sediments and persists for decades. Closely linked to legacy mining and industrial activity.

Nickel

Harmful at relatively low concentrations. Enters rivers mainly through mining runoff, industrial effluents, and urban stormwater.

River systems: Danube · Sava · Drina · Velika Morava · Južna Morava · Tisa · Tamiš · Timok

How the Method Works



From individual measurements → a spatial picture of where risk is high, why, and how it clusters

Four Ways to Look at Pollution Risk

Total Pollution Load



How much pollution is there?

The total amount of cadmium, lead, and nickel at each station, combined into one score. A simple but essential starting point — how heavy is the overall burden at this location?

Bioavailability



How dangerous is it to living things?

Not all pollution in the water is equally harmful. Some metals are locked in forms that fish and insects cannot absorb. This index measures only the fraction that can actually enter the food chain.

Trend Over Time



Is it getting better or worse?

How did metal levels change from 2019 to 2023? A location with moderate pollution that is steadily getting worse needs attention sooner than one with higher levels that are stable or improving.

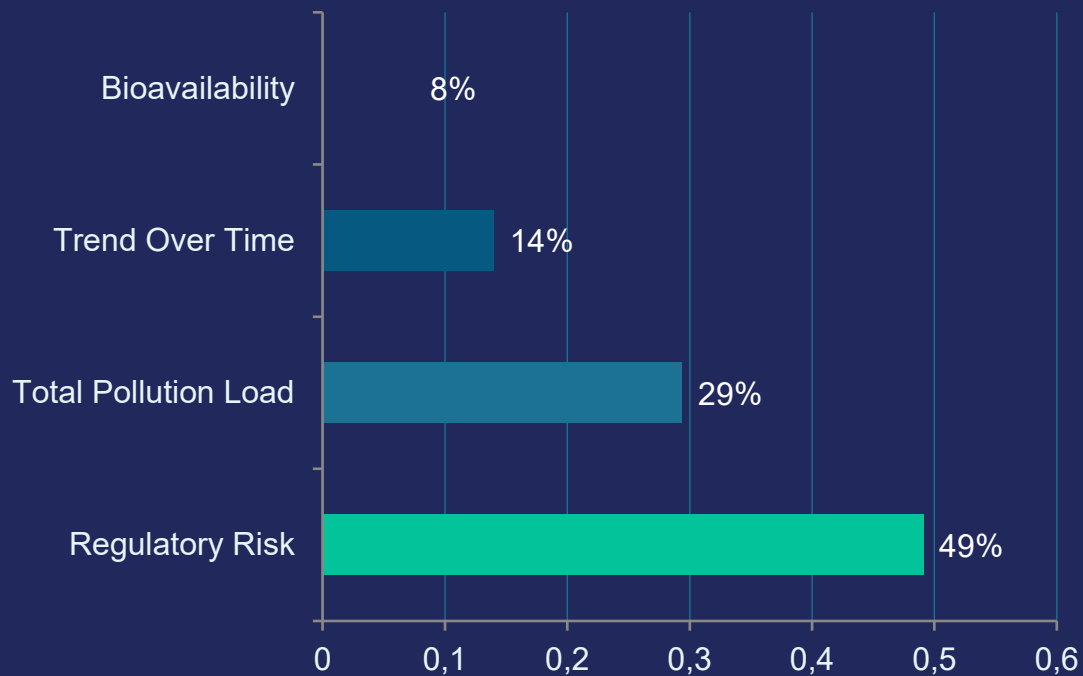
Regulatory Risk



How often are legal limits broken? • Weight: 49%

How frequently and how severely does each station exceed the limits set by Serbian and EU law? This carries the highest weight — because breaking legal limits is what triggers management action.

How I Weighted the Four Indexes (AHP)



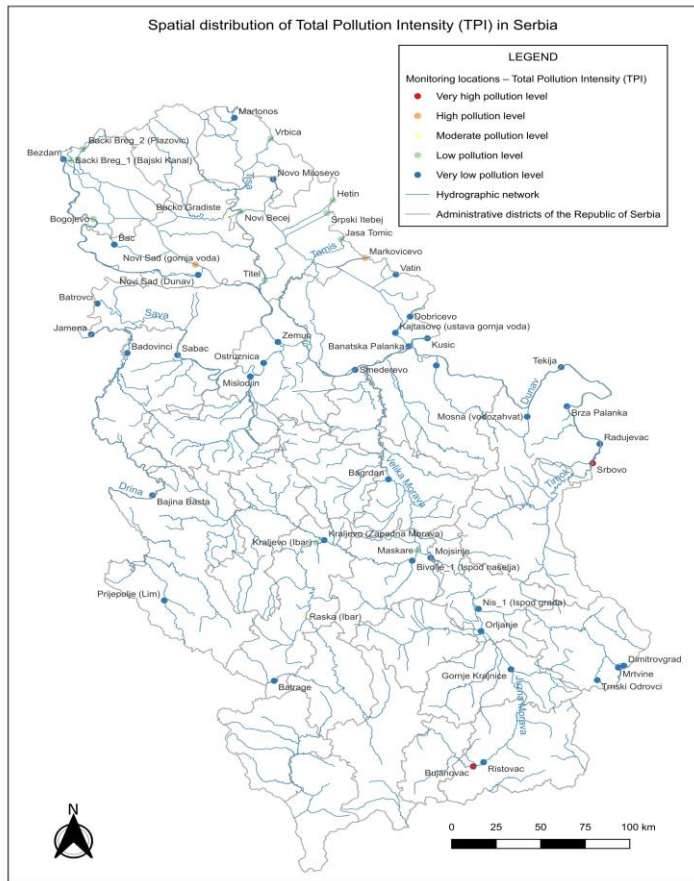
Regulatory risk carries the most weight —
breaking legal limits triggers real-world action.

Consistency Ratio = 0.022
(well below the 0.10 threshold)
Weights are logical and valid.

$IIR = 0.239 \times TPI + 0.086 \times BI$
 $+ 0.153 \times IT + 0.522 \times RRI$
Final risk score per station.

Total Pollution Load

How much pollution is there?



Five stations carry most of the burden

Srbovo, Bujanovac, Novi Sad (gornja voda), Markovićevo, and Bačko Gradište lead the ranking. The gap between these five and the rest of the network is striking.

Most stations are actually in good shape

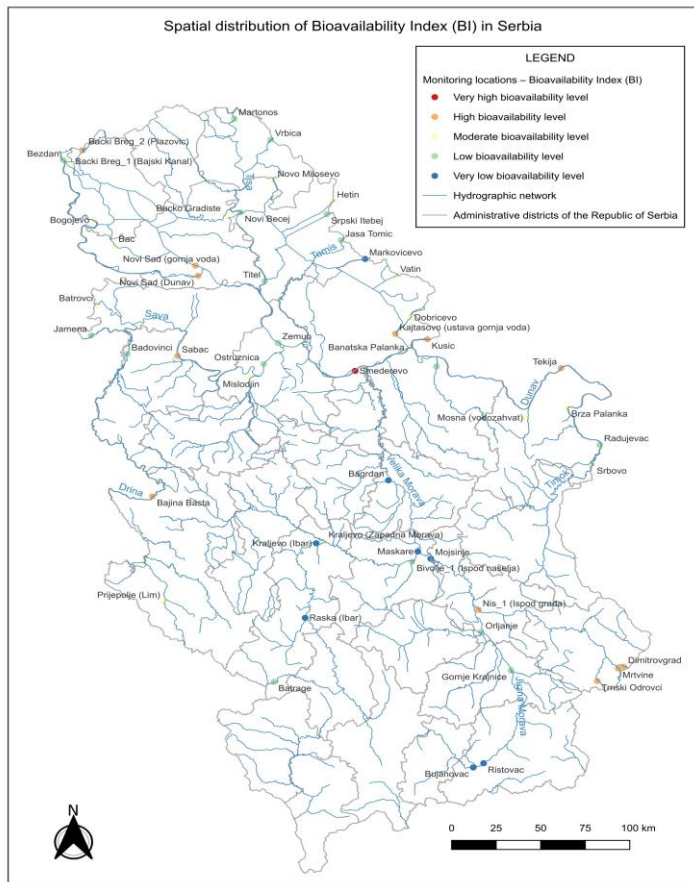
Most monitoring locations fall in the Very Low class. Heavy pollution is concentrated in just a few places, not spread evenly across the country.

North and south tell very different stories

High pollution levels in Vojvodina reflect diffuse agricultural loading. In the south, the highest values are linked to old mining areas.

Bioavailability

How dangerous is it to living things?



Smederevo stands far above the rest

Top five: Smederevo, Mrtvine, Tekija, Trnski Odrovci, Bački Breg 2. Smederevo's score is in a league of its own — dissolved metal fractions released near the steel plant.

High load does not always mean high danger

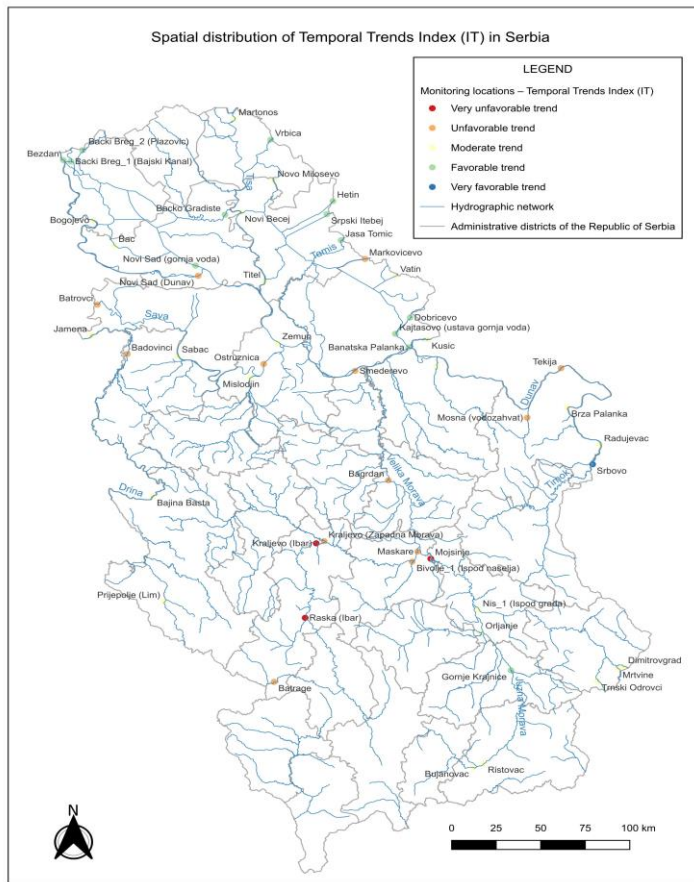
Several stations with moderate total pollution scores turn out to have high bioavailability. The total load alone would have missed this.

The Ibar catchment: contaminated but less ecologically active

Despite elevated metal concentrations, most metals along the Ibar are locked in stable sediment-bound forms that organisms cannot easily take up.

Trend Over Time

Is pollution getting better or worse?



The Ibar valley shows the strongest negative trend

Kraljevo (Ibar), Mojsinje, and Raska (Ibar) show the strongest negative trends, followed by Bagrdan and Maskare. All five showed acceptable levels in 2019. By 2023, the situation had changed significantly.

Srbovo is the one clear success story

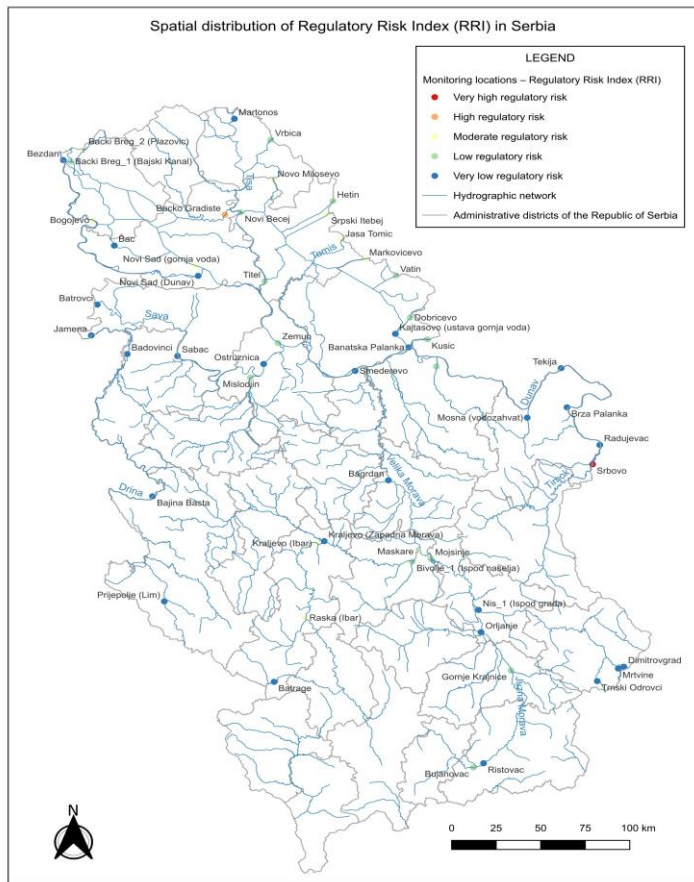
Despite the highest total pollution load in the network, Srbovo is the only station with a Very Favorable trend. Concentrations are coming down.

Most of the network is holding steady

More than half of all stations show moderate or favorable trends. The problem is not country-wide — it is concentrated in specific locations.

Regulatory Risk

How often are legal limits broken? · Weight: 49%



Srbovo has broken legal limits in almost every cycle

With a score of 1.0 — the maximum — Srbovo exceeded water quality standards in nearly every measurement over five years. Top five: Srbovo, Bačko Gradište, Raska (Ibar), Markovićevo, Jaša Tomić.

Two very different problems at the top

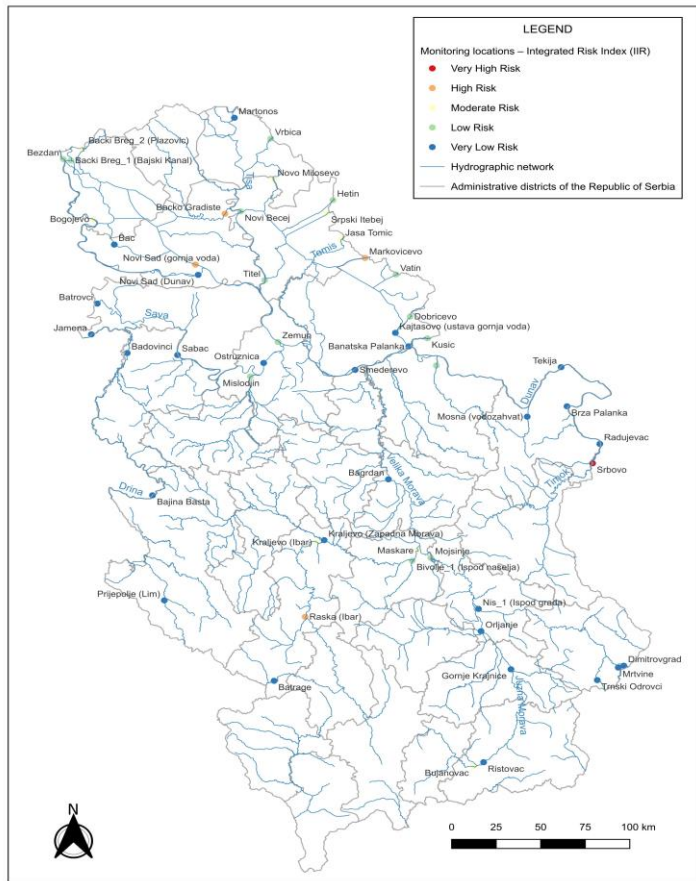
Srbovo's exceedances come from mining on the Timok. Bačko Gradište has persistent nickel problems from the canal network in Vojvodina. Same score, very different problem.

This index shapes the final risk map most strongly

Because regulatory risk carries 49% of the AHP weight, where it is high, the combined risk score follows.

Combined Risk Index (IIR)

All four factors together



This is the central result of the whole analysis

The final combined risk map — a single map that combines pollution load, ecological relevance, temporal behavior, and regulatory pressure.

Most of Serbia appears low-risk

A small number of locations clearly emerge as priority areas — and each one got there for a different reason.

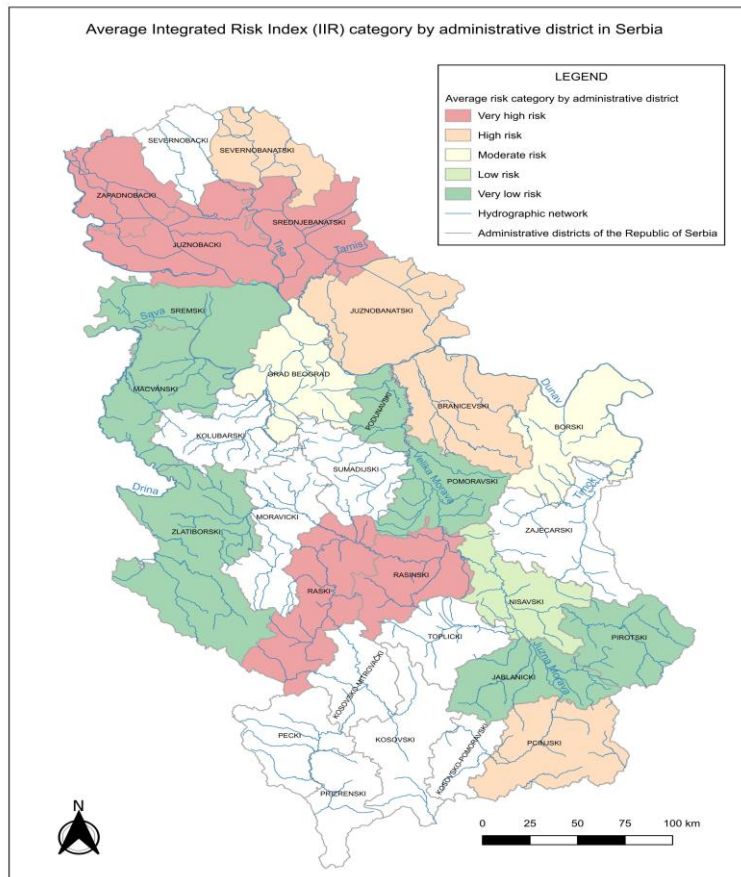
Each high-risk station got there a different way

Srbovo through mining and regulatory exceedances. Raska through a worsening trend. Smederevo through bioavailability.

Something no single index could have shown

This pattern only emerges when all four indexes are combined into one synthesized surface.

When Averages Hide the Problem



District averages can hide extreme local values.

The average has completely erased the worst station in the network.



Borski district — a clear example

The district average falls in the low risk category. Yet Srbovo, inside that district, has the highest Integrated Risk Index in the whole country.



Raška & Rasinski districts

Mining-related point sources dominate at the station level, but the district average smooths them into something that looks merely moderate.



Vojvodina (north)

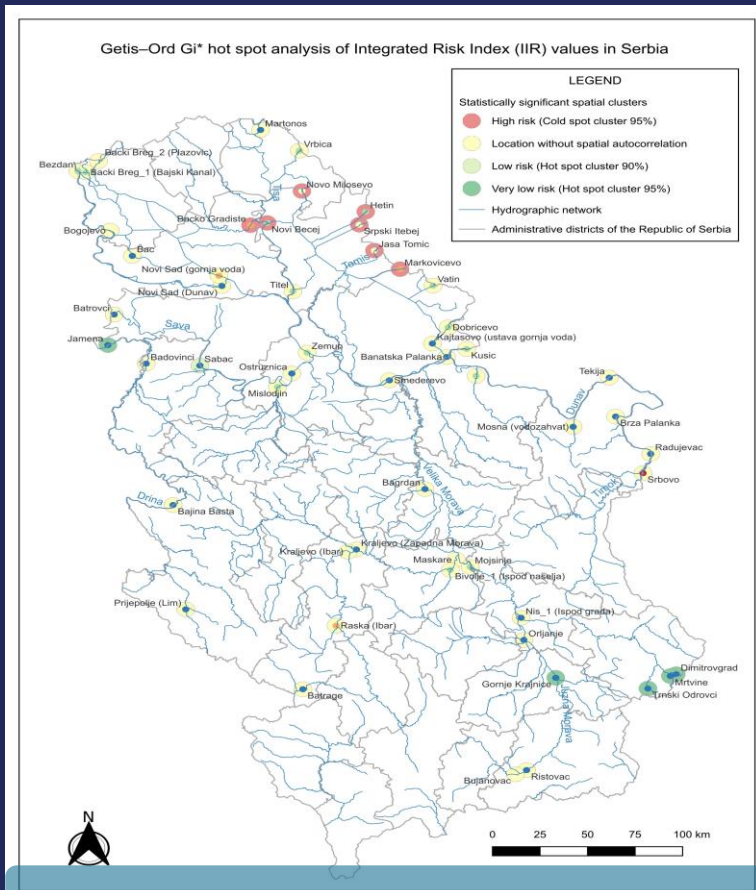
Agricultural loading raises the district average — while cleaner stations within the same area disappear into that same number.



The policy consequence

Environmental funding built on district averages will systematically miss the places that need help most.

Are High-Risk Stations Clustered in Space?



Cold Spots = High Risk Clusters

7 stations · Banat region · 95% confidence

Seven stations in the Banat share elevated risk AND their neighbors are also elevated. A regional problem driven by agriculture and industry — needs a regional response.

Hot Spots = Low Risk Clusters

5 stations · Southeast Serbia · 95% confidence

Five stations near Dimitrovgrad form the cleanest cluster in the network. Stable, uncontaminated conditions — valuable reference points for the whole monitoring system.

Spatial Outlier — Isolated Extreme

Srbovo · Very High risk · no high-risk neighbors

The most polluted station in Serbia — but surrounded by clean water. A local mining point source on the Timok. Needs a completely different response than the Banat cluster.

77.8% of stations show no significant spatial clustering — each one needs to be understood on its own terms

Why AHP + Getis-Ord Gi*?



AHP

“How risky is this location?”

Gives each station an absolute risk score — Class 1 to 5 — based on pollution load, bioavailability, trend, and regulatory risk.

Answers: WHERE is risk high?



Getis-Ord Gi*

“Does that risk form a pattern?”

Tells us whether a high-risk location is part of a wider cluster, or stands alone as an isolated case.

Answers: WHAT KIND of problem is it?

Together: not just WHERE risk is high, but WHAT KIND of problem it is — and what response it needs.

Assumptions, Constraints & Future Directions

Current Constraints



Network covers urban/industrial zones more than rural areas



Only 5 years — not enough for long-term patterns



Only 3 metals — other pollutants outside scope



AHP weights — expert-based, not statistically derived

Future Directions



Expand spatial coverage



Extend time series



Include more pollutant groups



Test weights & add ML prediction

These constraints do not invalidate the framework — they define its scope and give a clear roadmap for what comes next.

Open Source Perspective

Transparent

Every step is visible. There are no hidden calculations inside a proprietary black box.



QGIS

Reproducible

Anyone can take this workflow and run it themselves, and verify the results.

Adaptable

The same framework can be applied to other regions, other pollutants, other time periods.

Open science

Open-source GIS is not just a practical choice. It is a commitment to open, verifiable, and accessible science.

Conclusions



Environmental monitoring tells us what is happening.



Spatial analysis tells us where it matters.



And when we combine the two, we move beyond reporting environmental problems and begin to understand their spatial structure.



That understanding is essential for effective environmental management. And that, ultimately, is the value of GIS.

Thank you — Questions welcome

